INTRODUCTION

1. INTRODUCTION

1.1 SYSTEM OVERVIEW

LAWLY is a comprehensive legal assistance web application designed to simplify access to legal information and services for the common people of Kerala. The system provides an integrated digital environment where users can understand basic laws, consult registered advocates, handle legal documents, and complete essential legal tasks without requiring advanced legal knowledge. Built using Python Django and following the Model-View-Template (MVT) architectural pattern, LAWLY ensures smooth data flow, modular development, and efficient performance. HTML, CSS, and JavaScript form the interactive frontend, while SQLite3 serves as the lightweight and reliable backend database.

The system operates through three major modules—**User**, **Advocate**, and **Admin**—each equipped with tailored functionalities.

The **User module** is designed to offer maximum accessibility. Users can register, log in, browse advocate profiles, book appointments, view awareness content, search laws, create and analyse legal documents using predefined templates, make challan payments, check nearby legal offices, and access an emergency support feature. An AI-powered chatbot is integrated to provide instant responses to common legal queries, improving user guidance and understanding.

The **Advocate module** allows legal professionals to manage their profiles, interact with clients, view appointments, and assist with document-related tasks. However, certain features such as challan payment, awareness content browsing, and legal office viewing are restricted, ensuring role-based access control.

The **Admin module** oversees the entire system. Administrators can verify and approve registered advocates, add new laws, upload document templates, publish awareness content, manage user details, and monitor financial transactions like challan payments.

This ensures authenticity, security, and proper data management across the platform. Overall, LAWLY functions as an accessible legal ecosystem, providing reliable information, quick legal assistance, and simplified document handling. The system's goal is to enhance legal awareness, streamline advocate accessibility, and create a digital bridge between the public and essential legal services.

SYSTEM ANALYSIS

2. SYSTEM ANALYSIS

System development is a systematic process which includes phases such as planning, analysis, design, deployment, and maintenance. Systems Analysis is a process of collecting and interpreting facts, identifying the problems, and decomposition of a system into its components. System analysis is conducted for the purpose of studying a system or its parts in order to identify its objectives. It is a problem-solving technique that improves the system and ensures that all the components of the system work efficiently to accomplish their purpose. System will be analyzed in terms of their objectives, and the input, processes and output required achieving these objectives. The aims are to identify the boundaries of the system, the subsystem and the interfaces between subsystems.

To analyze a system means to study the system in detail. The analyst has to understand the functioning and concept of the system, before design the appropriate computer-based system that will meet all the requirements of the existing system. The system analyst has to carry out a customary approach to use the computer for problem solving.

System analysis involves studying the existing challenges faced by the target users and understanding how the proposed system can effectively solve these problems. For LAWLY, the analysis focuses on identifying legal awareness issues, accessibility barriers, document-related difficulties, and the need for a centralized digital legal platform.

2.1 EXISTING SYSTEM

In the current scenario, common people—especially those in Kerala—face numerous challenges when dealing with legal matters. There is a significant lack of legal awareness, as most individuals do not possess basic knowledge about essential rights, common laws, or legal procedures, leading to confusion, misinformation, and dependence on intermediaries. Accessing advocates is also difficult because no centralized platform exists to view advocate profiles, specializations, or contact details, making the process slow and unreliable. Legal document preparation adds another layer of difficulty, as documents such as affidavits, agreements, and declarations must be drafted manually by professionals, resulting in delays, extra costs, and repeated visits to legal offices. The challan payment process remains fully

manual, requiring individuals to visit government offices, fill out physical forms, stand in queues, and wait for verification. Additionally, there is no digital system to offer immediate legal support during emergency situations. Legal information, advocate details, law references, and awareness materials are scattered across multiple websites and unreliable sources, making it extremely hard for the public to find accurate and centralized legal guidance. As a result, the existing system is highly fragmented, time-consuming, and inaccessible for ordinary people.

2.2 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put for the with a very general plan for the proiect and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. Three key considerations involved in the feasibility analysis are:

- Economical feasibility
- Technical feasibility
- Operational feasibility
- Social feasibility

2.2.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have bon the organization. The amount of fund that the company can pour in to the research and development of the system is limited. The expenditures must be justified. Thus, the developed system as well with in the bud get and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

2.2.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is the technical requirements of the system. Any system developed must not that high demand on the available technical resources. This will lead to high demands on the available technical resources being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

2.2.3 OPERATIONAL FEASIBILTY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threat end by the system, instead must accept it as necessity. The level of acceptance by the user depends on the methods that are employed to educate the user about the system and to make him familiar with it.

2.2.4 SOCIAL FEASIBILTY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threat end by the system, instead must accept it as necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to makes constructive criticism, which is welcomed, as he is the final user of the system.

2.3 PROPOSED SYSTEM

The proposed system, LAWLY, is designed to address all major limitations of the existing legal support environment by providing a unified and user-friendly digital platform for the people of Kerala. LAWLY acts as a centralized legal assistance website where users can access legal information, connect with advocates, and understand essential legal procedures in one place. The platform features an intuitive interface built using HTML, CSS, and JavaScript, ensuring accessibility for users of all age groups. One of the key components of LAWLY is the AI-powered chatbot, which delivers instant responses to general legal queries, helping users gain quick awareness and guidance without professional intervention. The system also includes an advocate appointment module that allows users to browse advocate profiles, check their specializations and availability, and book appointments directly through the platform. To eliminate the inconvenience of manual government processes, LAWLY provides a digital challan payment facility, enabling users to pay their challans online with ease. Additionally, the platform supports automated document creation

and analysis through predefined templates, helping users generate legal documents quickly and accurately. For urgent situations, an emergency legal help feature is included, enabling users to request immediate assistance via a dedicated emergency button. LAWLY also ensures secure usage through role-based access control, offering separate modules for Users, Advocates, and Admin, each with specific permissions and functionalities. The Admin module plays a vital role in maintaining system integrity by approving advocate registrations, adding laws, uploading document templates, managing awareness content, viewing payment details, and supervising the overall activity on the platform. Through these comprehensive features, the proposed system provides a highly efficient, accessible, and centralized solution to support the legal needs of the public service alerts for their next service dates. It is developed to minimize the manual process.

2.3.1 MODULE/DESCRIPTION

I. USER

- Profile: View and edit profile, change password.
- Advocate Services: Browse advocate profiles, search advocates, and book appointments.
- Legal Information: Search laws, view legal awareness content, and understand basic legal rights.
- Documents: Create, analyze, and download legal documents using predefined templates.
- Feedback: Reply to feedback given by users.
- Payments: Make challan payments and view payment history
- Support: AI-powered chatbot for legal queries

II. ADVOCATE

- Profile: Create, view, and manage advocate profile details
- Appointments: View user bookings and manage appointment schedules.
- Law: View law and search law

III. ADMIN

- User Management: View and manage user and advocate details.
- Advocate Verification: Verify and approve registered advocates.
- Law Management: Add, update, and manage laws in the system.
- Content Management: Upload document templates and publish legal awareness content.
- Payments: Monitor challan payments and financial transactions.
- System Control: Ensure security, authenticity, and proper data management

2.4 REQUIREMENT ANALYSIS

In the study phase, which is the first step in the development of this new system, I tried to understand how the existing system works, its disadvantages, and whether a new system that provides additional features to the existing system can be designed. The requirement analysis involves the identification of objectives and requirements, evaluation of alternative solutions, and recommendation of a more feasible solution. In other words, requirement analysis is a systematic process of gathering, recording, and implementing facts.

In the study phase of my project, I analyzed and identified most of the problems of the existing system and their limitations and thought of whether a new system with new features can be designed This includes the identification of objectives and requirements, evaluation of alternative solutions and recommendation for more feasible solution. Here, I analyzed and identified most of the problems of the existing system and tried my level best to collect as much of information necessary for this project.

SYSTEM SPECIFICATION

3. SYSTEM SPECIFICATION

System specification means the essential facilities for working of our new application software. There are mainly two classifications for system specification. They are:

- Software Specification
- Hardware Specification

3.1 SOFTWARE SPECIFICATION

The major element in building a system is the selection of compactable software requirement of the system on which the project was developed is as follows:

- Operating System: Microsoft windows 7 or above
- Front End: HTML, JavaScript, CSS
- Framework: Django
- Programming Language: Python
- Back End: sqlite3
- IDE: VS code 2022
- Web Browser: Google Chrome, Microsoft Edge

3.2 HARDWARE SPECIFICATION

The selection of hardware configuration is a very important task related to the software development. The processor should be powerful to handle entire operation.

- Hard Disk : 150 GB.
- Processor: Intel® Core™ i3 or above
- RAM: 4 GB RAM minimum, 8GB recommended

SYSTEM DESIGN

4. SYSTEM DESIGN

The most creative and challenging phase of the system development is system design, is a solution to how to approach to the creation of the proposed system. It refers to the technical specification that will be applied. It provides the understanding and procedural details necessary for implementing the system recommended in the feasibility study. Design goes through the logical and physical stages of development. At an early stage in designing a new system, the system analyst must have a clear understanding of the objectives, which the design is aiming to fulfill. The different steps of system designer:

- Input design
- Output design
- Database design
- Normalization
- Data Flow Diagram (DFD)
- E-R Diagram

4.1 INPUT DESIGN

Input design is a part of overall system design, which requires careful attention. It is the process of converting user-originated inputs to a computer- based format. The major objective of the input design is to make data entry easy, logical and error free.

A form is "any surface on which information is to be entered, the nature of which is determined by what is already on the surface". If the data going into the system is incorrect, then processing and output will magnify these errors. So, designer should ensure that form is acceptable and understandable by the user. This application has been developed in a user-friendly manner. The layout of the form is made in such a way that the user will not find any difficulty in going from one field to other by just pressing the tab. During the processing the cursor is placed in the position where the data must been entered.

The user is also provided with an option of selecting an appropriate input from a list of values. Necessary dropdown list boxes and combo boxes are included for necessary

fields so that the user need not remember all the data and can just select from it.

Validation is made for each and every data entered. Help messages are also provided whenever the users enter a wrong data into a particular field. This makes the user to understand what is to be entered, moreover whenever an erroneous data is entered the error message is displayed and the user can move to the next field only after entering the correct data. The clear label for the menus and fields is also provided. Consultations are provided so that a user can view the details of any process at any time.

4.2 OUTPUT DESIGN

One of the most important features of a system for users in the output it produces. Output design should improve the system's relationship with the user and help in decision-making. Considering the future use of output required, and depending on the nature, it is displayed on the monitor for immediate need of obtaining the hard copy. The objective of output design is to define the controls and format of all printed documents and reports and of screens that will be produced by the system. Computer output is the most important and direct source of information to the user.

Objective of output design :

- Design output to serve the intended purpose.
- Deliver the appropriate quality of output.
- Choose the right output method.
- Provide output on time.

4.3 DATABASE DESIGN

The activity deals with the design of the data base. A key is to determine how the paths are to be implemented. A physical path is derived from a logical path. The general theme behind database is to handle information as a whole.

A database is a collection of interrelated data stored with a minimum of redundancy

to serve many applications. It minimizes the artificiality embedded in using separate files. The designer determines what data must be stored and how the data elements interrelate with this information, they can begin to fit the data to the database model. The organization of data in a database aims to achieve three major objectives. They are

- In a database information from several files are coordinated, accessed and operated upon as though it is a single file.
- In a database all data are stored in the place only and it allows each application to access it. This approach results in more consistent information.
- To structure the data so that any pertinent relationship between entities can be represented.
- To permit simple retrieval of data in responses.

4.4 NORMALIZATION

Normalization is the process of breaking up the database into smaller tables to obtain higher consistency and efficiency and to reduce redundancy. It is the main tool used for designing tables in a database. The fields in each table are normalized by this concept. After normalization we will get a number of tables containing set of fields with minimum redundancy and interdependency. During the process of normalization, we can identify dependencies, which can cause problems when deleting or updating details. Normalization also helps to simplify the structure of tables. A fully normalized table consist of:

- Primary key that identifies an entity.
- Set of attributes other than the primary key, which are functionally dependent on the primary key, describes the entity.

4.4.1 FIRST NORMAL FORM (1 NF)

A table is said to be in First Normal Form if :

- The domain of each attribute contains only atomic values
- The values of each attribute contain only a single value from that domain

4.4.2 SECOND NORMAL FORM (2 NF)

A table is said to be in Second Normal Form if :

- It is in INF and
- No non-prime attribute is dependent on any proper subset of any candidate key

TABLES

5. TABLES

USER

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	id	Primary key	Integer	user
2	fullname	Not null	Varchar(200)	Full name of the user
3	email	Not null	Email/ Varchar(200)	User email address
4	language	Not null	Varchar(200)	Preferred language of the user
5	location	Not null	Varchar(100)	User location
6	contact_no	Not null	Varchar(200)	User contact number
7	password	Not null	Varchar(200)	User login password

LAW

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	law_id	Primary Key	Integer (Auto Increment)	Unique law ID
2	category	Not Null	Varchar(100)	Category of the law
3	section_no	Null	Varchar(100)	Section number of the law
4	title	Not Null	Varchar(100)	Title of the law
5	description	Null	Text	Detailed description of the law
6	sourcelink	Null	URL / Varchar	Source or reference link
7	punishment	Null	Text	Punishment details
8	bailable_status	Null	Varchar(100)	Indicates whether the offence is Bailable or Non-Bailable
9	cognizable_status	Null	Varchar(100)	Indicates whether the offence is Cognizable or Non-Cognizable
10	triable_by	Null	Varchar(100)	Authority or court that can try the case

LEGAL_TEMPLATE

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	template_id	Primary Key	Integer	Unique template ID
2	template_name	Not Null	Varchar(255)	Name of the legal template
3	category	Not Null	Varchar(100)	Category of the template
4	description	Null	Text	Brief description of the template
5	template_content	Not Null	Text	Template body with placeholders
6	uploaded_file	Null	File	Uploaded legal template document
7	created_at	Null	Datetime	Date and time of template creation

FEEDBACK

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	id	Primary key	Integer	Unique feedback ID
2	user	Foreign key	Integer	Reference to use data
3	advocate	Foreign key	Integer	Reference to advocate
4	rating	Not null	Integer	Rating given by the user.
5	message	Not null	Text	User feedback message
6	created_at	Not null	Datetime	Date and time of feedback submission

ADVOCATE

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	advocate_id	Primary key	Integer	Unique advocate ID
2	advocate_name	Not null	Varchar(255)	Full name of the advocate
3	specialization	Not null	Varchar(255)	Area of legal specialization
4	email	Not null	Email	Advocate email address
5	password	Not null	Varchar(255)	Advocate login password
6	experience_year	Not null	Integer	Years of professional experience
7	location	Not null	Varchar(255)	Advocate practice location
8	availability_status	Null	Boolean	Availability for consultation
9	state_bar_council	Null	Varchar(255)	Registerd state bar council
10	enrollment_number	Null	Varchar(255)	Bar council enrollment number
11	bar_council_idcard	Null	File	Uploaded bar council id card
12	image	Null	Image	Advocate profile photo
13	qualification	Null	Varchar(255)	Academic and professional qualifications
14	is_verified	Null	Boolean	Admin verification status
15	verified_at	Null	Datetime	Date and time of admin approval

PAYMENT

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	transaction_id	Primary key	Integer	Unique transaction ID
2	user_id	Foreign key	Integer	Reference to use data
3	challan_no	Not null	Varchar(100)	Challan number for which payment is made
4	amount	Not null	Decimal	Amount paid for the challan
5	status	Not null	Boolean	Payment status
6	payment_date	Not null	Datetime	Date and time of payment

AWARENESS_CONTENT

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	content_id	Primary key	Integer	Unique awareness content ID
2	title	Not null	Varchar(200)	Title of the awareness content
3	type	Not null	Varchar(200)	Type of content
4	link	Not null	Varchar(200)	Link to the awareness content
5	upload_date	Not null	Datetime	Date and time of content upload
6	category	Not null	Varchar(200)	Category of legal awareness content

LEGAL_OFFICE

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	office_id	Primary key	Integer	Unique legal office ID
2	office_name	Not null	Varchar(200)	Name of the legal office
3	address	Not null	Text	Complete office address
4	location	Not null	Varchar(200)	City of the legal office
5	contact_no	Not null	Varchar(20)	Contact number of the legal office
6	email	Not null	Email	Official email address
7	offences_handled	Not null	Text	Types of offences handled
8	working_hours	Not null	Varchar(200)	Office working hours

APPOINTMENT

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	appointment_id	Primary key	Integer	Unique appointment ID
2	user	Foreign key	Integer	Reference to userData
3	advocate	Foreign key	Integer	Reference to advocate
4	appointment_date	Not null	Date	Scheduled date of appointment
5	status	Not null	Varchar(50)	Appointment status

CHALLAN

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	challan_no	Primary key	Integer	Unique challan number
2	user	Foreign key	Integer	Reference to userData
3	offence_type	Not null	Varchar(200)	Type of offence commited
4	offence_location	Not null	Varchar(200)	Location where offence occurred
5	issue_date	Not null	Date	Date on which challan was issued
6	due_date	Not null	Date	Last date to pay the challan
7	fine_amount	Not null	Decimal	Fine amount imposed
8	description	Not null	Text	Detailed offence description
9	status	Not null	Boolean	Payment status

LEGAL_DOCUMENT

SI.NO	FIELD NAME	CONSTRAINTS	DATATYPE	DESCRIPTION
1	id	Primary key	Integer	Unique identifier for each legal document record
2	user	Foreign key	Integer	Links the document to the user who uploaded it
3	file	Not null	Integer	Uploaded legal document file
4	uploaded_at	Not null	Datetime	Stores the date and time when the document was uploaded
5	ai_summary	Not null	Text	Ai generated summary of the uploaded legal document

DATA FLOW DIAGRAM

6. DATA FLOW DIAGRAM

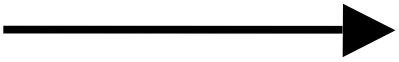
A data flow diagram (DFD) is a diagram that describes the flow of and the processor that change data throughout system. It's a structured analysis ad design tool that can be used for flow charting in place of or in association with information.

Oriented and process-oriented system flowcharts. When analysts prepare the data flow diagram, they specify the user needs at a level of detail that virtually determines the information flow into and out of the system and the required data resources. This network is constructed by using a set of symbols that do not imply physical implementations.

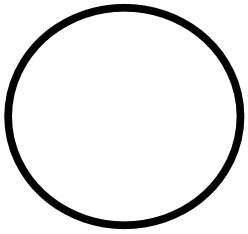
The data flow diagram reviews the current physical system, prepares input and output specification, Specifies the implementation plan etc. Four basic symbols are used to construct data flow diagrams. They are symbols that represent data source, data flows, and data transitions and data storage. The points at which data are transformed are represented by enclosed figures, usually circles, which are called nodes.

Data flow diagram symbols:

Source or destination of data



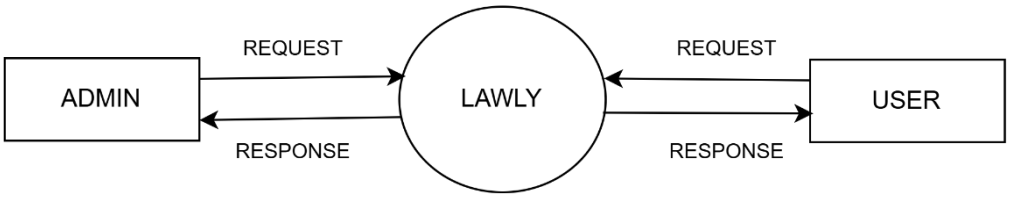
Data Flow



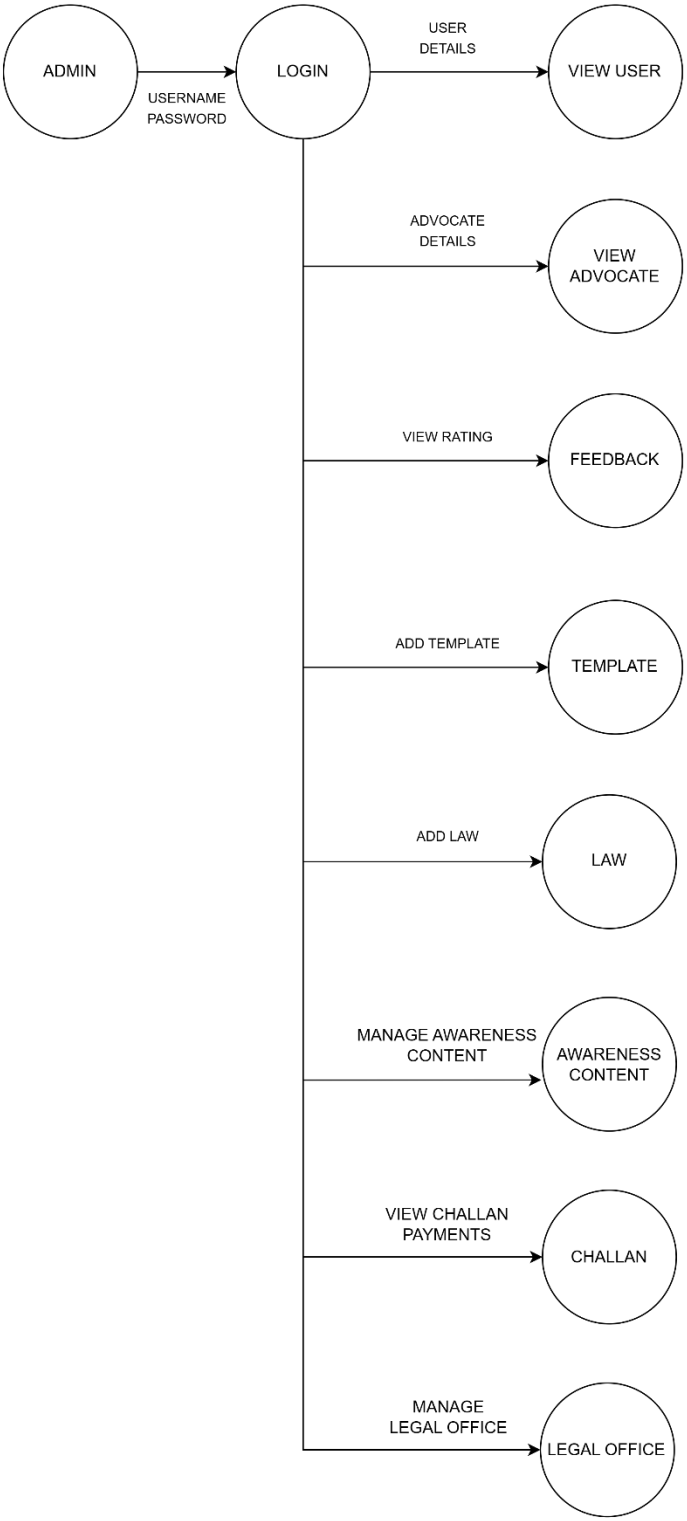
Process

Storage

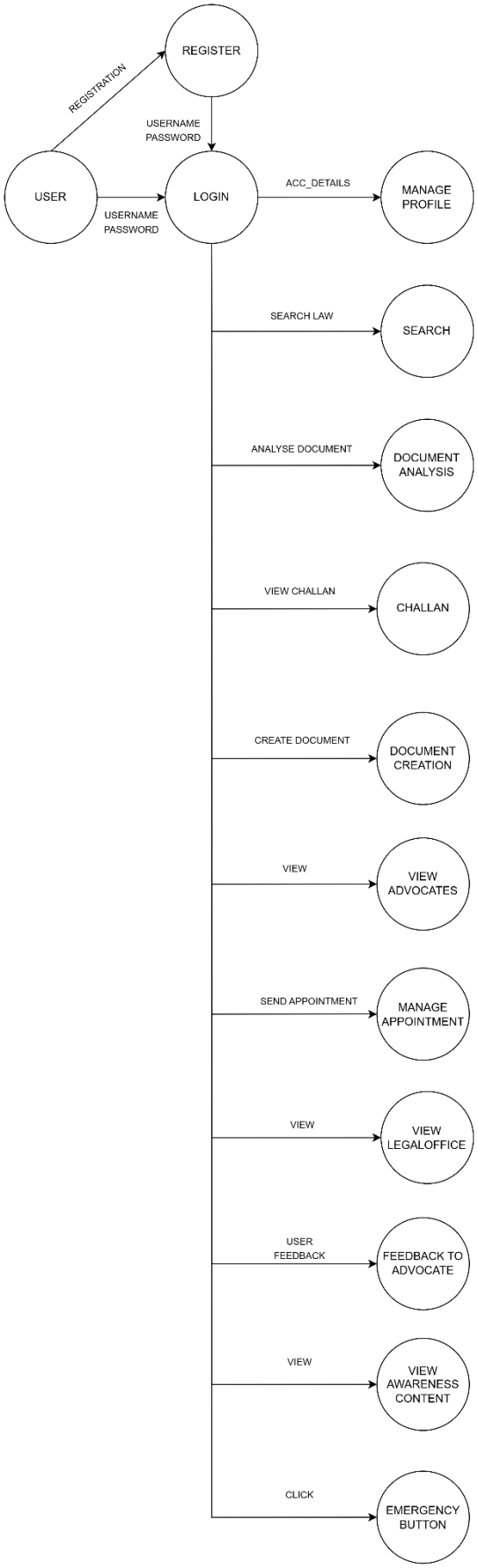
LEVEL : 0



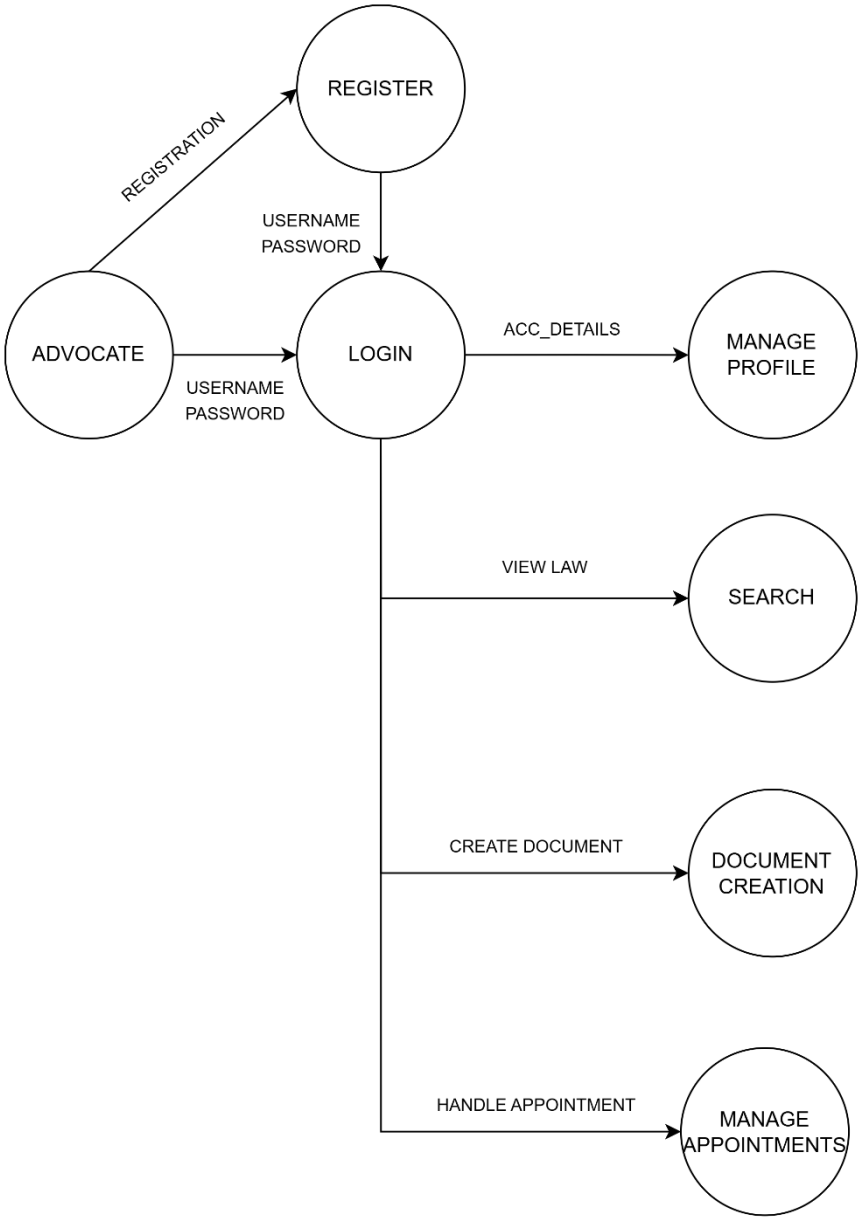
LEVEL 1 : ADMIN



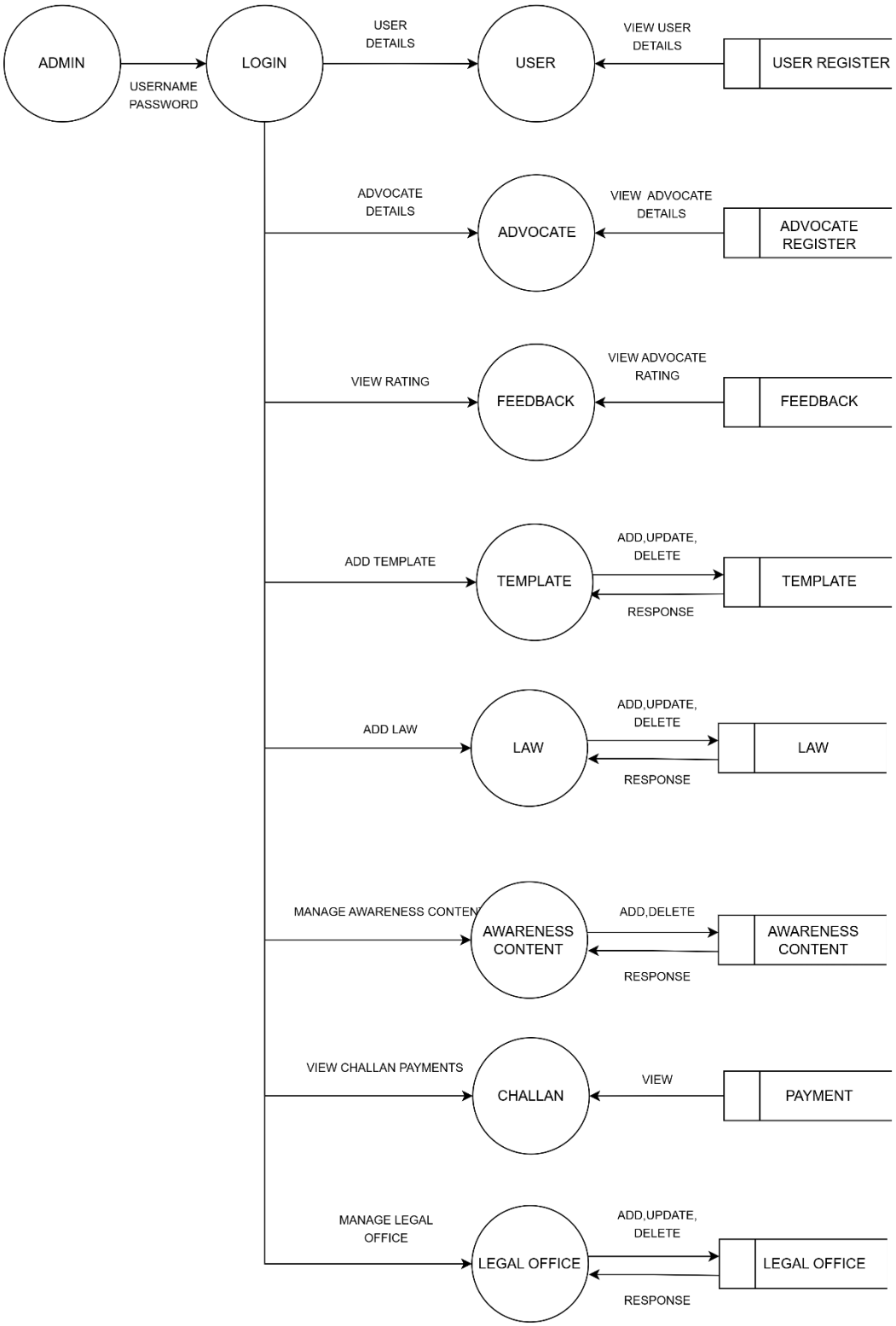
LEVEL 1 : USER


REGISTER
REGISTRATION
USERNAME
PASSWORD
USER
LOGIN
USERNAME
PASSWORD
ACC_DETAILS
MANAGE PROFILE
SEARCH LAW
SEARCH
ANALYSE DOCUMENT
DOCUMENT ANALYSIS
VIEW CHALLAN
CHALLAN
CREATE DOCUMENT
DOCUMENT CREATION
VIEW
VIEW ADVOCATES
SEND APPOINTMENT
MANAGE APPOINTMENT
VIEW
VIEW LEGALOFFICE
USER FEEDBACK
FEEDBACK TO ADVOCATE
VIEW
VIEW AWARENESS CONTENT
CLICK
EMERGENCY BUTTON

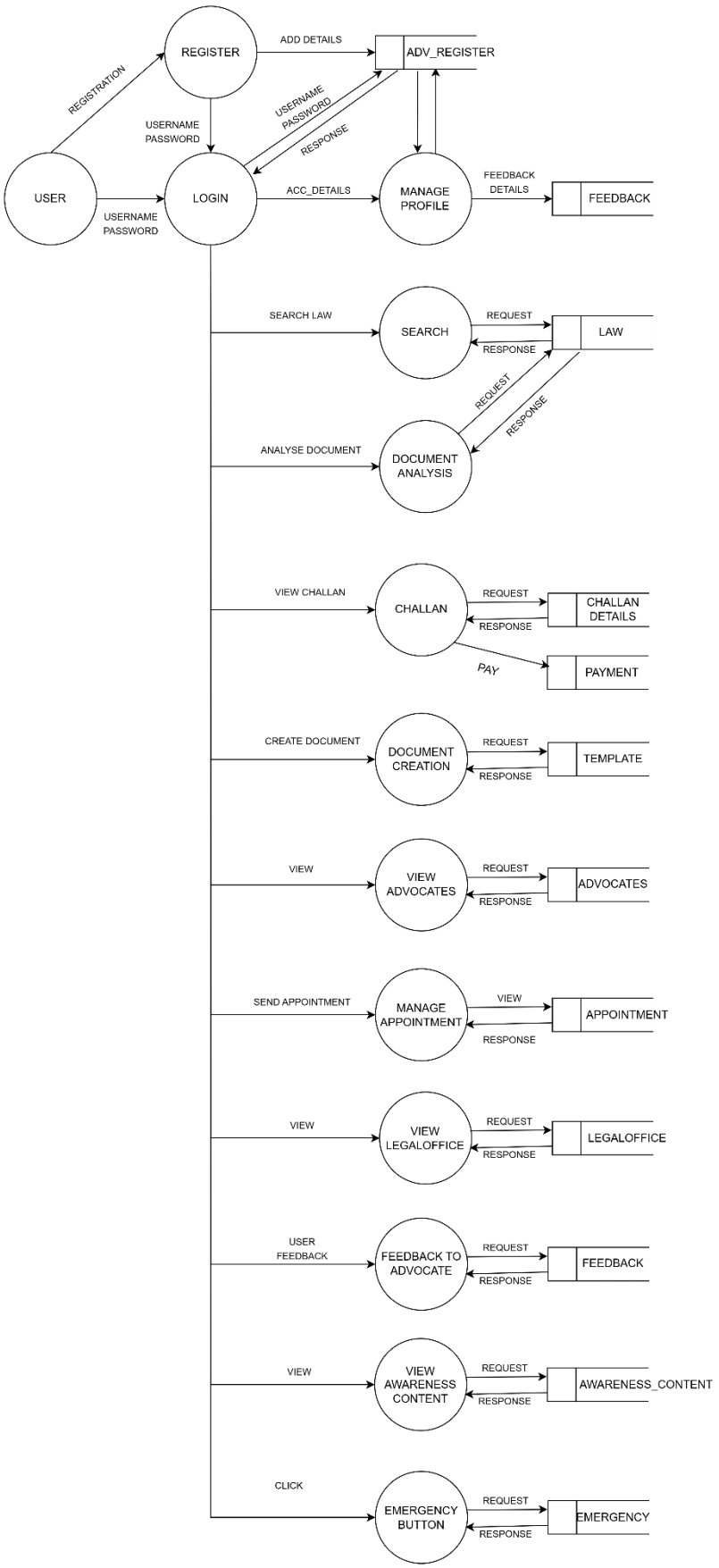
LEVEL 1: ADVOCATE



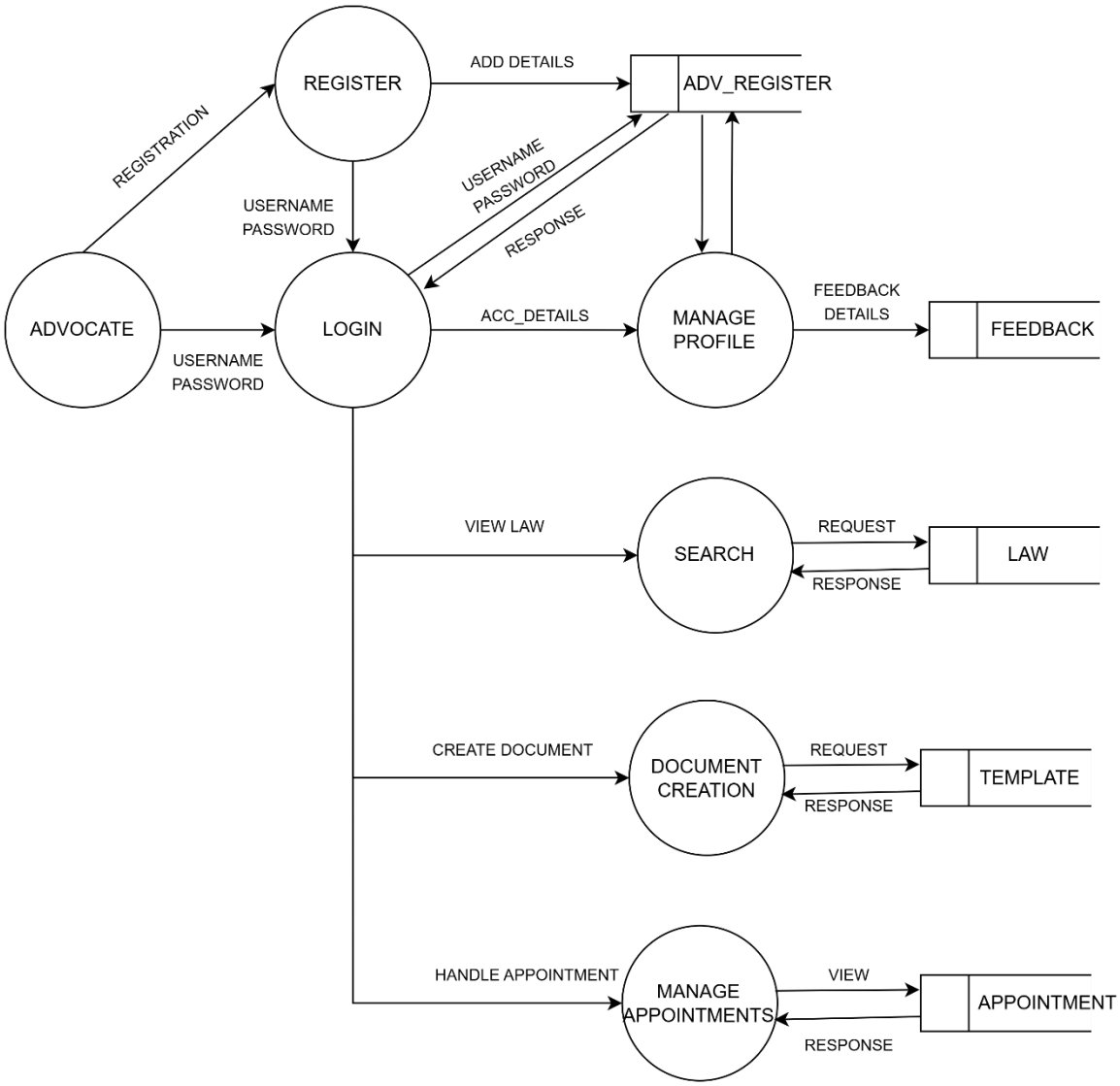
LEVEL 2: ADMIN



LEVEL 2 : USER

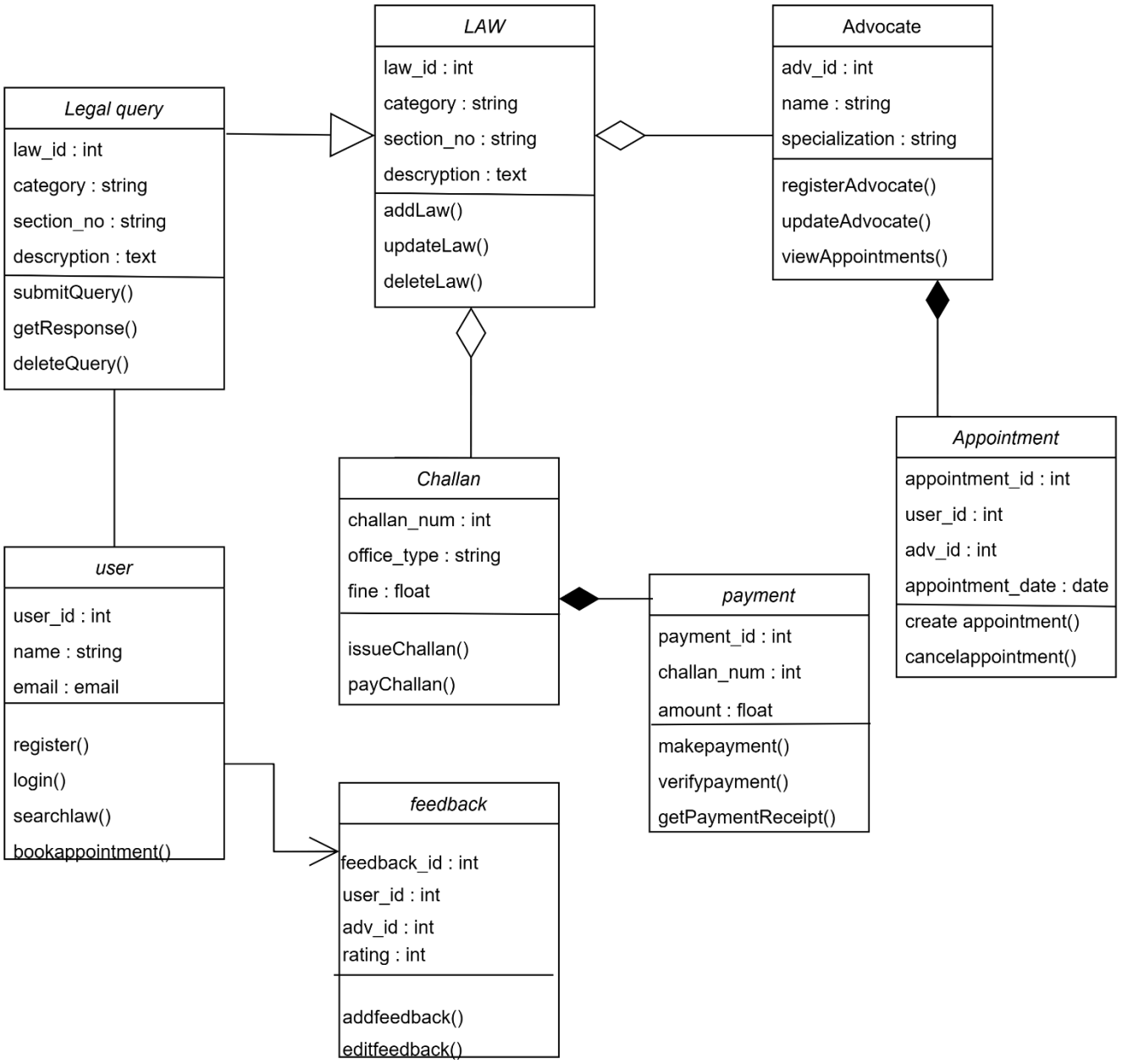

REGISTRATION
REGISTER
ADD DETAILS
ADV_REGISTER
USERNAME
PASSWORD
USERNAME
PASSWORD
RESPONSE
USER
USERNAME
PASSWORD
LOGIN
ACC_DETAILS
MANAGE
PROFILE
FEEDBACK
DETAILS
FEEDBACK
SEARCH LAW
SEARCH
REQUEST
RESPONSE
LAW
REQUEST
RESPONSE
ANALYSE DOCUMENT
DOCUMENT
ANALYSIS
VIEW CHALLAN
CHALLAN
REQUEST
RESPONSE
CHALLAN
DETAILS
PAY
PAYMENT
CREATE DOCUMENT
DOCUMENT
CREATION
REQUEST
RESPONSE
TEMPLATE
VIEW
VIEW
ADVOCATES
REQUEST
RESPONSE
ADVOCATES
SEND APPOINTMENT
MANAGE
APPOINTMENT
VIEW
RESPONSE
APPOINTMENT
VIEW
VIEW
LEGALOFFICE
REQUEST
RESPONSE
LEGALOFFICE
USER
FEEDBACK
FEEDBACK TO
ADVOCATE
REQUEST
RESPONSE
FEEDBACK
VIEW
VIEW
AWARENESS
CONTENT
REQUEST
RESPONSE
AWARENESS_CONTENT
CLICK
EMERGENCY
BUTTON
REQUEST
RESPONSE
EMERGENCY

LEVEL 2 : ADVOCATE



CLASS DIAGRAM

7. CLASS DIAGRAM

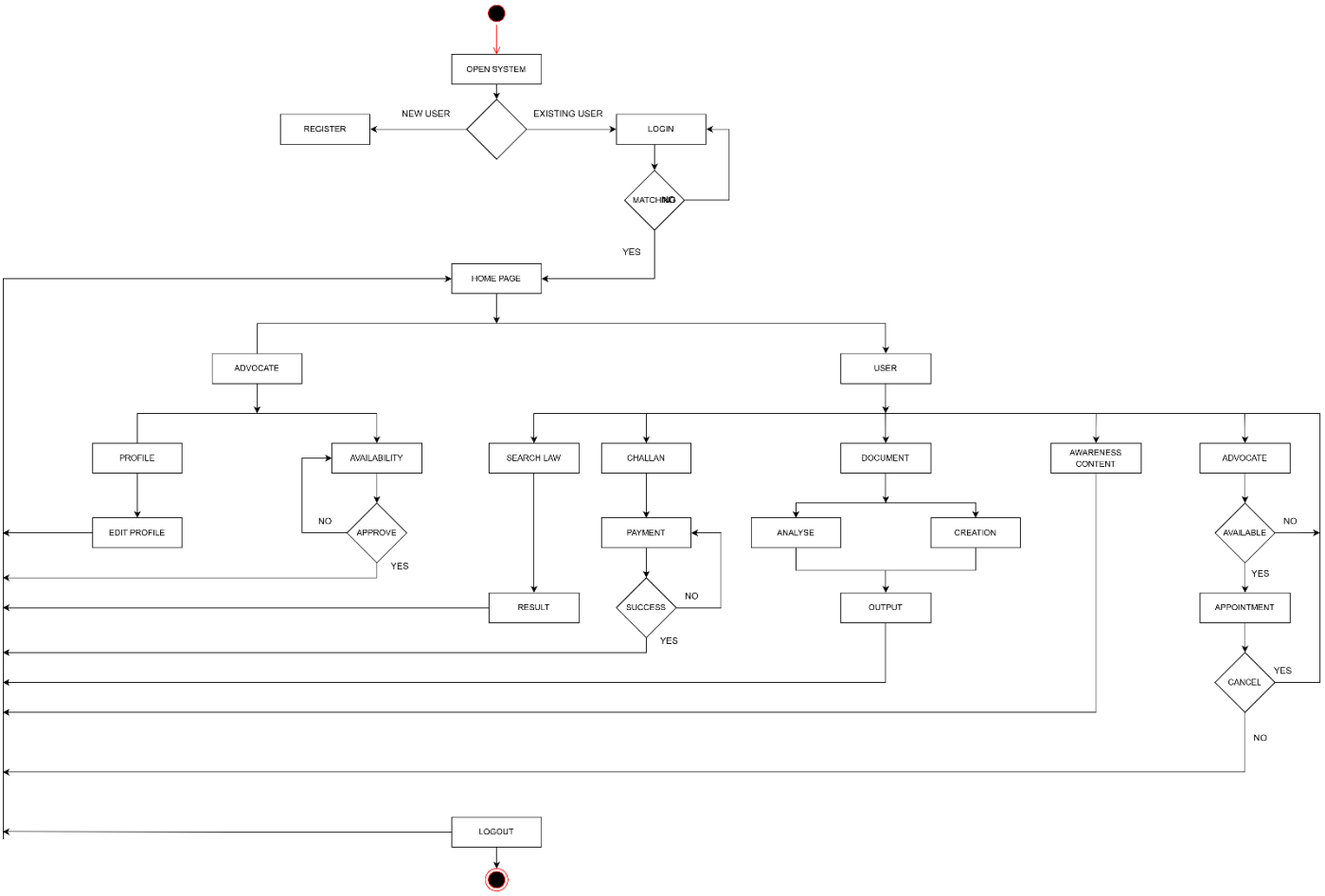


USECASE DIAGRAM

8. USECASE DIAGRAM

ACTIVITY DIAGRAM

9. ACTIVITY DIAGRAM



CODE DESIGN

Code design refers to the process of planning and creating the structure, organization, and logic of computer code for a software project. It involves defining the architecture, components, interfaces, and algorithms that make up the code base, as well as selecting the appropriate programming language and tools for the task at hand.

Good code design is important because it can make code more efficient, easier to maintain, and more scalable over time. A well-designed codebase can also make it easier for multiple developers to collaborate on a project and can reduce the likelihood of bugs and errors.

Code design typically involves a combination of top-down and bottom- up approaches. Top-down design involves starting with the high-level architecture and working down to the individual components and algorithms, while bottom-up design involves starting with the individual code components and building up to the higher-level architecture. Both approaches are important for creating a cohesive and effective codebase.

HTML

The Hypertext Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript. Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document. HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as and directly introduce content into the page. Other tags such as surround and provide information about document text and may include other tags as sub-elements. Browsers do not display the HTML tags but use them to interpret the content of the page.

HTML can embed programs written in a scripting language such as JavaScript, which affects the behavior and content of web pages. Inclusion of CSS defines the look and layout

of content. The World Wide Web Consortium (W3C), former maintainer of the HTML and current maintainer of the CSS standards, has encouraged the use of CSS over explicit presentational HTML since 1997.A form of HTML, known as HTML5, is used to display video and audio, primarily using the element, in collaboration with JavaScript.

JAVASCRIPT

JavaScript often abbreviated JS, is a programming language that is one of the core technologies of the World Wide Web, alongside HTML and CSS. Over 97% of websites use JavaScript on the client side for web page behavior, often incorporating third-party libraries. All major web browsers have a dedicated JavaScript engine to execute the code on users' devices. JavaScript is a high-level, often just-in-time compiled language that conforms to the ECMAScript standard. It has dynamic typing, prototype-based object orientation, and first-class functions. It is multi-paradigm, supporting event driven, functional, and imperative programming styles. It has website programming interfaces (APIs) for working with text, dates, regular expressions, standard data structures, and the Document Object Model (DOM).

The ECMAScript standard does not include any input/output (I/O), such as networking, storage, or graphics facilities. In practice, the web browser or other runtime system provides JavaScript APIs for I/O. JavaScript engines were originally used only in web browsers, but are now core components of some servers and a variety of websites. The most popular runtime system for this usage is Node.js. Although Java and JavaScript are similar in name, syntax, and respective standard libraries, the two languages are distinct and differ greatly in design.

CSS

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML.[1] CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript. CSS is designed to enable the separation of presentation and content, including layout, colors, and fonts. This separation can improve content accessibility; provide more flexibility and control in the specification of presentation characteristics; enable multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, which reduces complexity and repetition in the structural content; and enable the .css file to be cached to

improve the page load speed between the pages that share the file and its formatting. Separation of formatting and content also makes it feasible to present the same markup page in different styles for different rendering methods, such as on-screen, in print, by voice (via speech-based browser or screen reader), and on Braille based tactile devices. CSS also has rules for alternate formatting if the content is accessed on a mobile device. The name cascading comes from the specified priority scheme to determine which style rule applies if more than one rule matches a particular element. This cascading priority scheme is predictable. The CSS specifications are maintained by the World Wide Web Consortium (W3C). Internet media type MIME type) text/css is registered for use with CSS by RFC 2318 (March 1998). The W3C operates a free CSS validation service for CSS documents. In addition to HTML, other markup languages support the use of CSS including XHTML, plain XML, SVG, and XUL.

PYTHON

Python, one of the most popular programming languages in the world, has created everything from Netflix's recommendation algorithm to the software that controls self-driving cars. Python is a general-purpose language, which means it's designed to be used in a range of applications, including **data science, software and web development, automation**, and generally getting stuff done. Python is commonly used for developing websites and software, task automation, data analysis, and data visualization. Since it's relatively easy to learn, Python has been adopted by many non-programmers such as accountants and scientists, for a variety of everyday tasks, like organizing finances.

Python is often used to develop the back end of a website or application the parts that a user doesn't see. Python's role in web development can include sending data to and from servers, processing data and communicating with databases, URL routing, and ensuring security. Python offers several frameworks for web development. Commonly used ones include Django and Flask.

Some web development jobs that use Python include back end engineers, full stack engineers, Python developers, software engineers, and DevOps engineers.

SQLITE

SQLite is an in-process library that implements a self-contained, serverless, zero-configuration, transactional SQL database engine. It is a popular choice as an embedded database for local/client storage in application software such as web browsers. It is also used in many other applications that need a lightweight, embedded database. SQLite is ACID-compliant and implements most of the SQL standards, using a dynamically and weakly typed SQL syntax that does not guarantee domain integrity.

To use SQLite in a C/C++ program, you can use the sqlite3 API, which provides a lightweight, simple, self-contained, high-reliability, full- featured, and SQL database engine. The API is implemented as a library of C functions that can be called from your program. One of the main benefits of using SQLite is that it is very easy to get started with. To create a new database in SQLite, you simply need to create a new file on your filesystem and connect to it using the sqlite3 API.

SYSTEM TESTING

10.SYSTEM TESTING

The most important phase in system development life cycle is system testing. The number and nature of errors in a newly designed system depends on the system specifications and the time frame given for the design. A newly designed system should have all the subsystems working together, but in reality, each subsystem works independently. During this phase all the subsystems are gathered into one pool and tested to determine whether it meets the user requirements.

Testing is done at two levels- Testing of individual modules and testing the entire system. During the system testing, the system is used experimentally to ensure that the software will run according to the specification and in the way the user expects. Each test case is designed with the intent of finding errors in the way the system will process it.

Testing plays a very critical role in determining the reliability and efficiency of software and hence is a very important stage in software development. Software testing is done at different levels:

- Unit Testing
- System Testing
- Integration Testin
- Acceptance Testing

10.1 UNIT TESTING

This is the first level of testing. The different modules are tested against the specification produced during the integration. This is done to test the internal logic of each module. Those resulting from the interaction between modules are initially avoided. The input received and output generated is also tested to see whether it fails in the expected range of values. Unit testing is performed from bottom up, starting with the smallest and lowest modules and proceeding one at a time. The programs are tested for correctness of logic applied and detection of errors in coding. Each of the modules was tested and errors are rectified. The license provider has been done by the unit testing.

10.2 SYSTEM TESTING

The objective of system testing is to check if the software meets its requirements. System testing is done to uncover errors that were not found in earlier tests. This includes forced system failures and validation of total system as the user in the operational environment implements it. Under this testing low volumes of transaction are generally based on live data. The total system is also tested for recovery after various major failures to ensure that no data are lost during the breakdown.

10.3 INTEGRATION TESTING

In integration testing the tested modules are combined into sub-systems, which are then tested. The goal of integration testing to check whether the module can be integrated properly emphasizing on the interface between modules. The software protector module is linked together and integration testing done on them.

10.4 ACCEPTANCE TESTING

The objective of acceptance testing is to tell the user about validity and reliability of the system. It verifies whether the system operates as specified and the integrity of important data is maintained. User motivation is very important for the successful performance of the system.

10.5 TESTING FOR THE PROPOSED SYSTEM

Testing a purpose-built system involves evaluating the system's ability to meet its intended purpose or objective. This involves verifying that the system functions as intended and that it delivers the desired results. Here are some key steps involved in testing a purpose-built system:

- Define the system's purpose and objectives: The first step is to clearly define the system's purpose and objectives. This involves understanding the requirements and expectations of the stakeholders and end-users.

- Develop test cases: Test cases are developed to ensure that the system functions as intended and that it meets the requirements and objectives. Test cases should cover all aspects of the system, including functional and non- functional requirements.
- Execute test cases: The test cases are executed to verify that the system meets the requirements and objectives. This includes testing the system's functionality, usability, performance, and security.
- Report and track issues: Any issues or defects identified during testing should be reported and tracked to ensure that they are resolved before the system is deployed.
- Retest: Once the issues have been resolved, the system should be retested to ensure that the fixes have been effective and that the system meets the requirements and objectives.
- Obtain stakeholder and end-user approval: Once the testing is complete, the system should be reviewed and approved by the stakeholders and end- users to ensure that it meets their expectations and requirements.

Overall, testing a purpose-built system involves verifying that the system meets its intended purpose and that it delivers the desired results. This requires a thorough understanding of the system's requirements.

SYSTEM IMPLEMENTATION

AND

MAINTENANCE

11. SYSTEM IMPLEMENTATION AND MAINTENANCE

It is the stage of implementation, which ensures that system works accurately and effectively before the live operation commences. It is a confirmation that all are correct and opportunity to show the users that the system must be tested with text data and show that the system will operate successfully and produce expected result under expected conditions.

Before implementation, the proposed system must be tested with raw data to ensure that the modules of the system work correctly and satisfactorily. The system must be tested with valid data to achieve its objective.

The purpose of system testing is to identify and correct errors in the candidate system. As important as this phase is, it is one that is frequently compromised. Typically, the project the schedule or the user is eager to go directly to conversion. Actually, testing is done to achieve the system goal. Testing is vital to the parts of the system are correct, the goal will be successfully achieved. Inadequate testing or non-testing leads to errors that may not appear until months later.

This creates two problems:

- The time lag between the cause and appearance of the problem.

- The effect of system errors on files and records within the system. A small system errors can conceivably explode into much larger problem. Effectively early in the process translates directly into long term cost saving from a reduced number of errors.

In our project we properly installed our system for our client in the following way…

- Hardware configuration installed properly.
- Front-end and back-end software installed properly
- Path settings are corrected.
- Run the software

11.1 SYSTEM MAINTENANCE

After a system successfully implemented, it should be maintained in a proper manner. System is performing to make the system adaptable to the changing environment. Maintenance has 3 types:

- Corrective Maintenance
- Adaptive Maintenance
- Perfective Maintenance

11.2 CORRECTIVE MAINTENANCE

Software testing will not user uncover all the errors in the system. During the use of any large program errors will occur and reported to the developer. The process that includes the correction of errors called corrective maintenance.

11.3 ADAPTIVE MAINTENANCE

It is an activity that modifies the entire system to properly inter with the changing environment.

11.4 PERFECTIVE MAINTENANCE

It is performed to enhance the system by making modification in the program to meet the user's changing needs.

FUTURE ENHANCEMENT

12. FUTURE ENHANCEMENT

Although LAWLY provides a strong and effective platform for legal assistance, there are several opportunities for future enhancements that can further improve its functionality, user experience, and overall impact. One potential enhancement is the integration of multilingual support, especially in Malayalam, to make the system more accessible to a wider audience in Kerala. Advanced AI features such as predictive legal suggestions, case outcome analysis, and automated document review can be implemented to provide more intelligent guidance to users. The platform can also be extended to include online consultation sessions through video conferencing, enabling real-time interaction between users and advocates. Introducing a secure e-signature facility would simplify document verification and further reduce the need for physical visits. The system could incorporate GPS-based tracking to locate nearby police stations, courts, or legal offices during emergencies.

A mobile application version of LAWLY can also be developed for easier access and convenience. Integration with government portals, payment gateways, and court case status APIs can enhance reliability and automation. Additionally, a notification system that alerts users about new laws, case updates, appointment reminders, and challan deadlines can improve user engagement. As the platform grows, implementing enhanced security features such as two-factor authentication, data encryption, and role-based permission upgrades will ensure better protection of user data. Overall, these future enhancements can transform LAWLY into a more intelligent, user-centric, and comprehensive legal support ecosystem.

CONCLUSION

13. CONCLUSION

LAWLY is a comprehensive and user-friendly legal assistance platform designed to bridge the gap between common people and the legal system. By addressing the major challenges of the existing environment—such as lack of legal awareness, difficulty in accessing advocates, slow document preparation, and non-digital challan processes—the system provides an efficient and modern solution tailored for the people of Kerala. Through its integration of features like an AI-powered chatbot, advocate appointment scheduling, digital challan payment, automated document creation, and emergency legal support, LAWLY offers a centralized, accessible, and highly functional platform that simplifies legal procedures for everyday users. The use of Python Django with the MVT architecture, along with HTML, CSS, JavaScript, and SQLite3, ensures that the system is secure, scalable, and easy to maintain.

The role-based structure involving Users, Advocates, and Admin further enhances operational transparency and control. Overall, LAWLY succeeds in creating a digital ecosystem that promotes legal awareness, improves service accessibility, and empowers individuals to handle legal matters with greater confidence and convenience. It stands as a meaningful contribution toward modernizing and democratizing legal services in Kerala.

SCREEN LAYOUT

14. SCREEN LAYOUT

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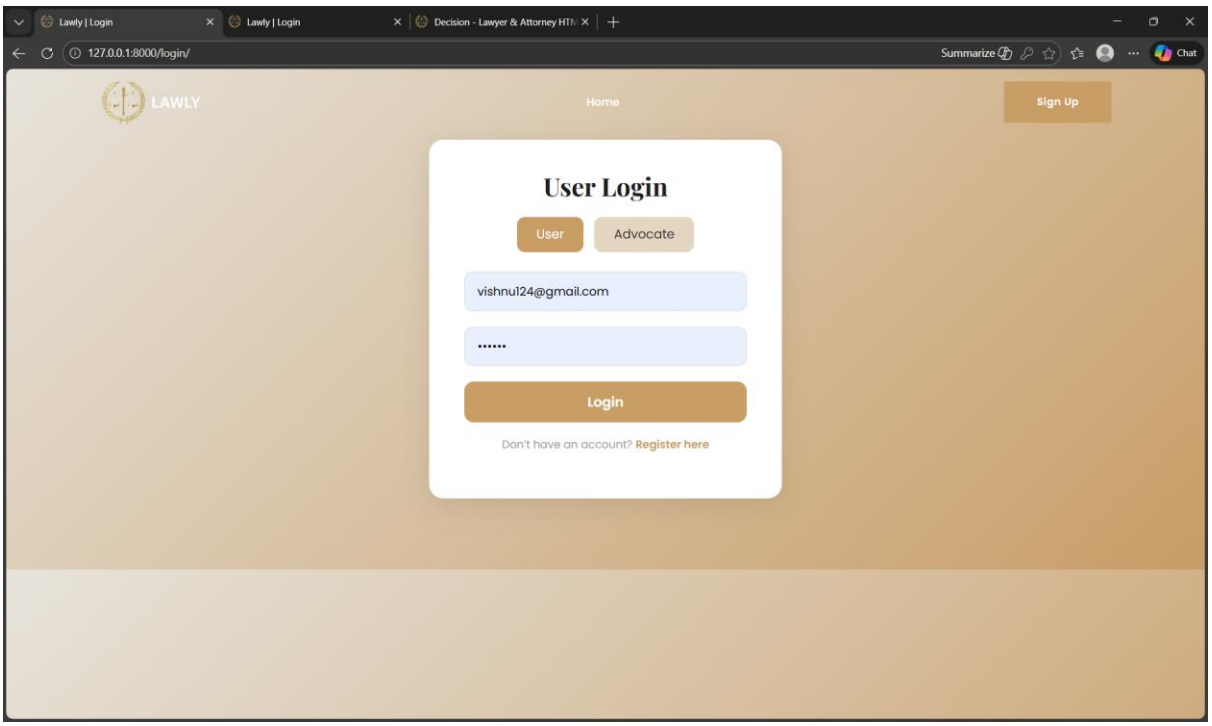

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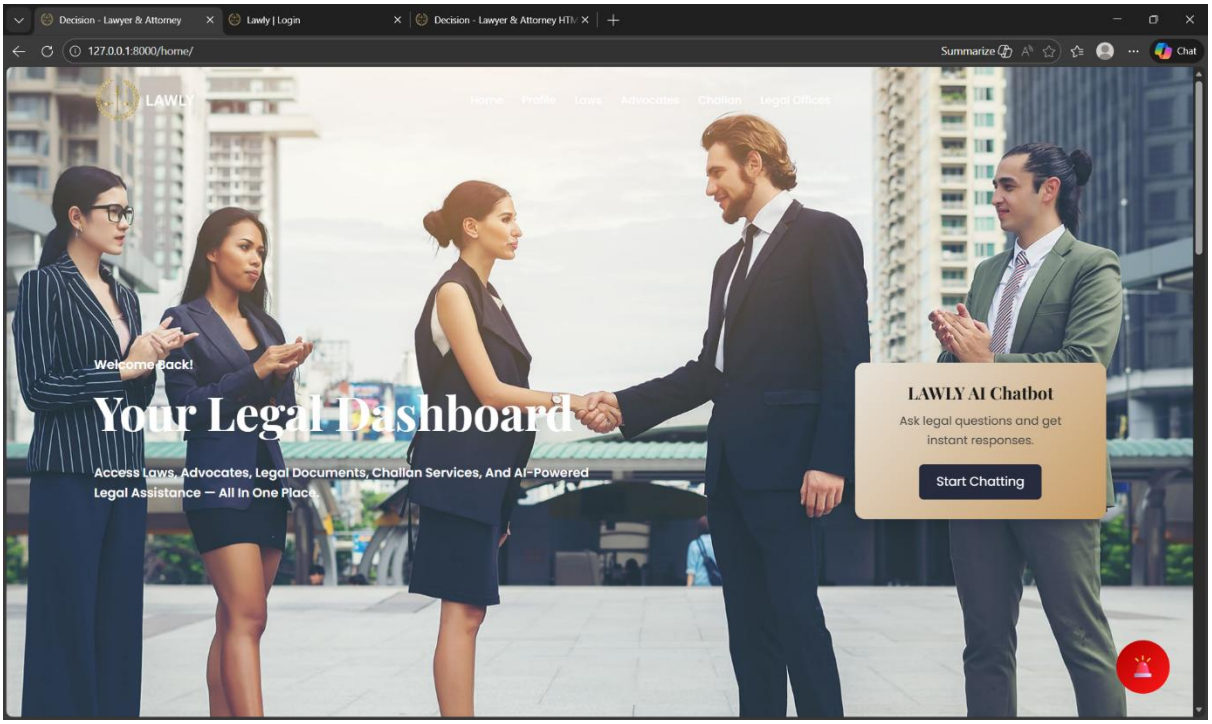

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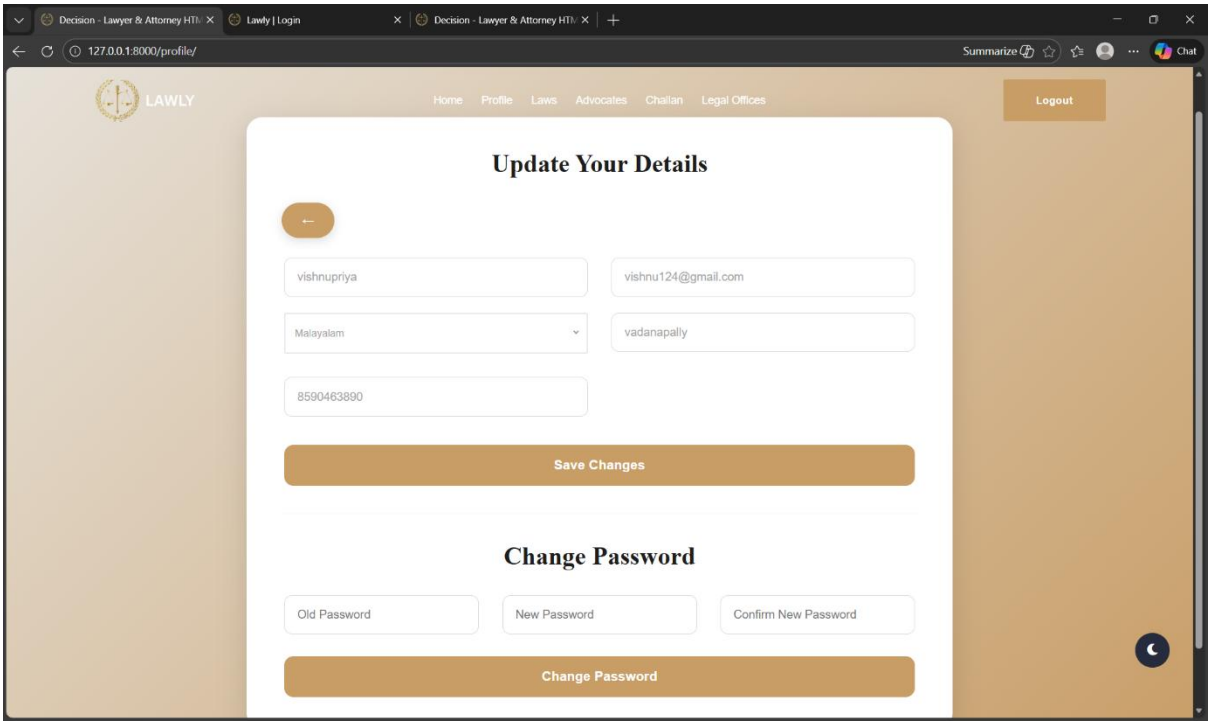

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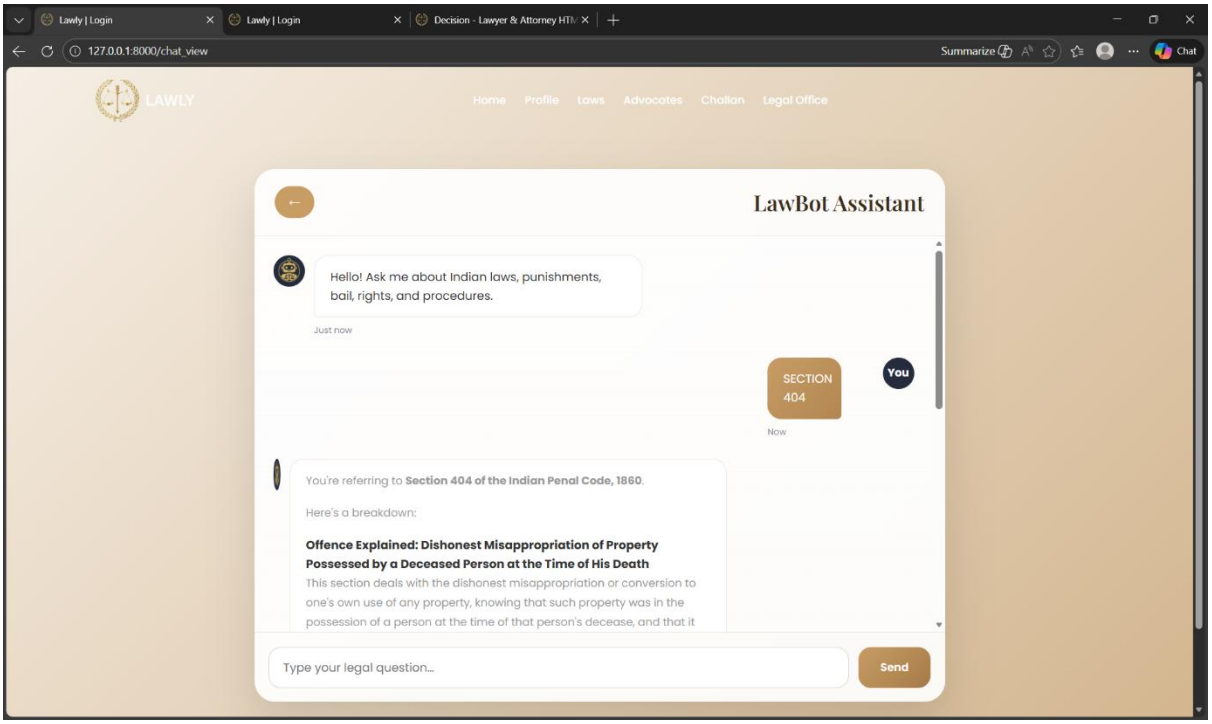

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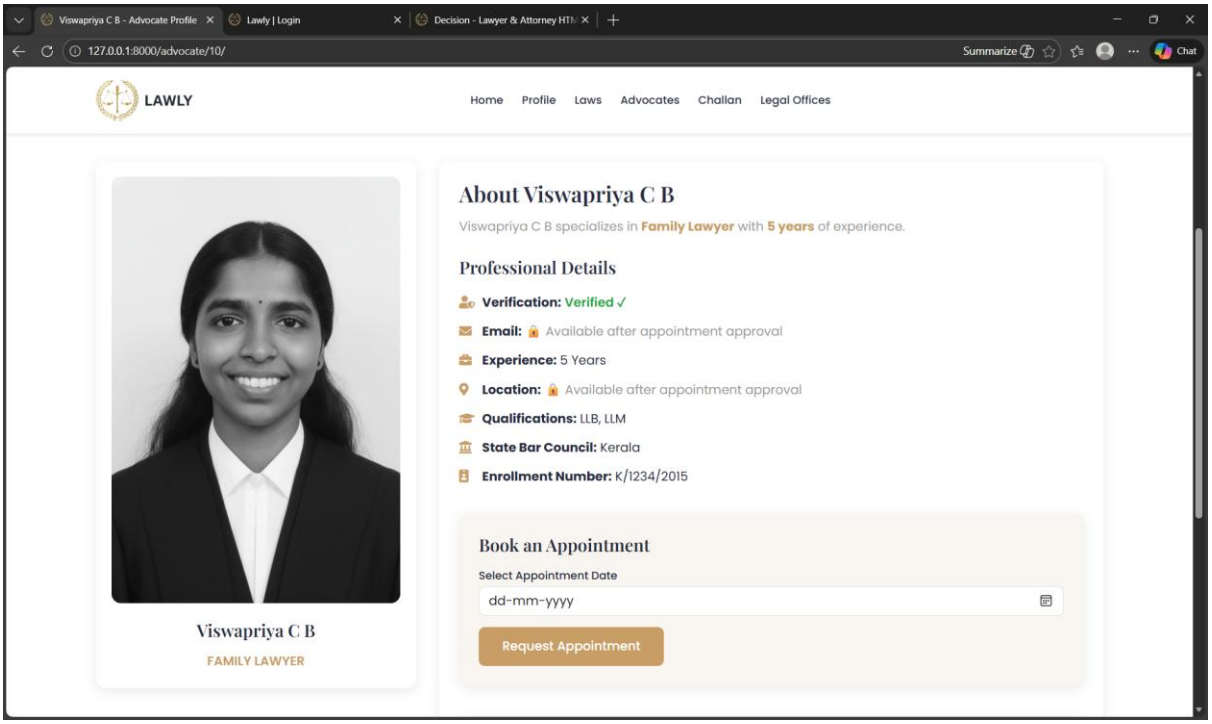

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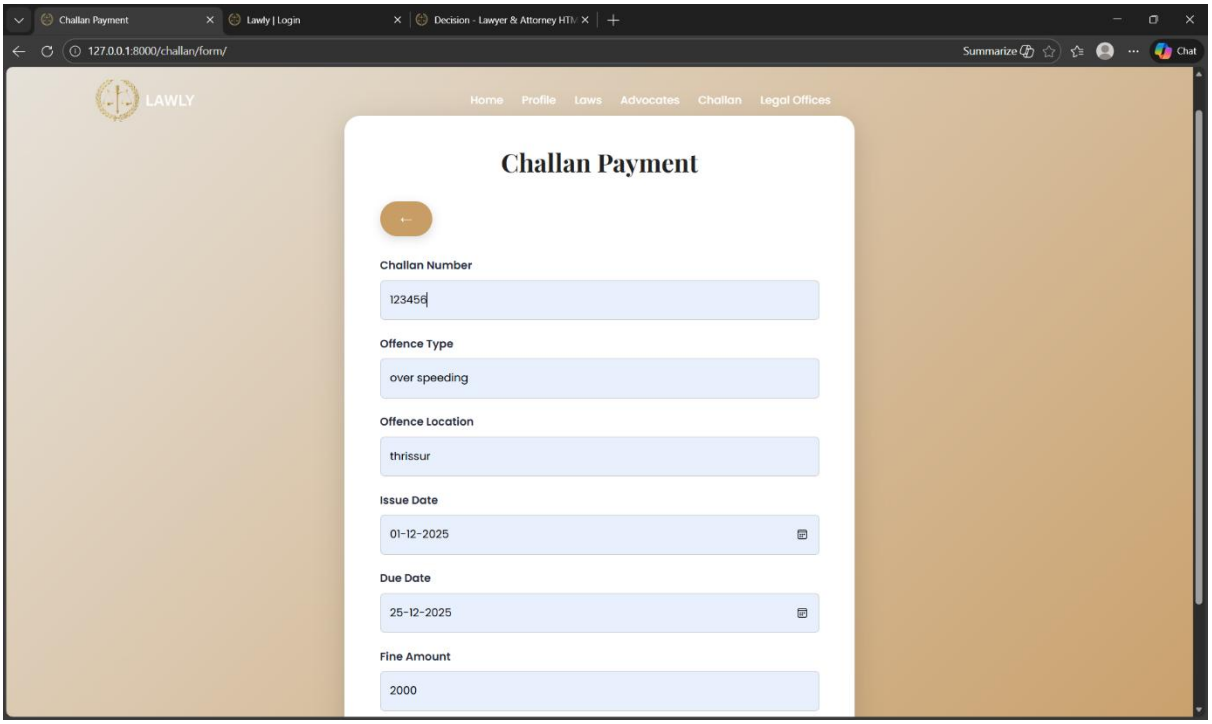

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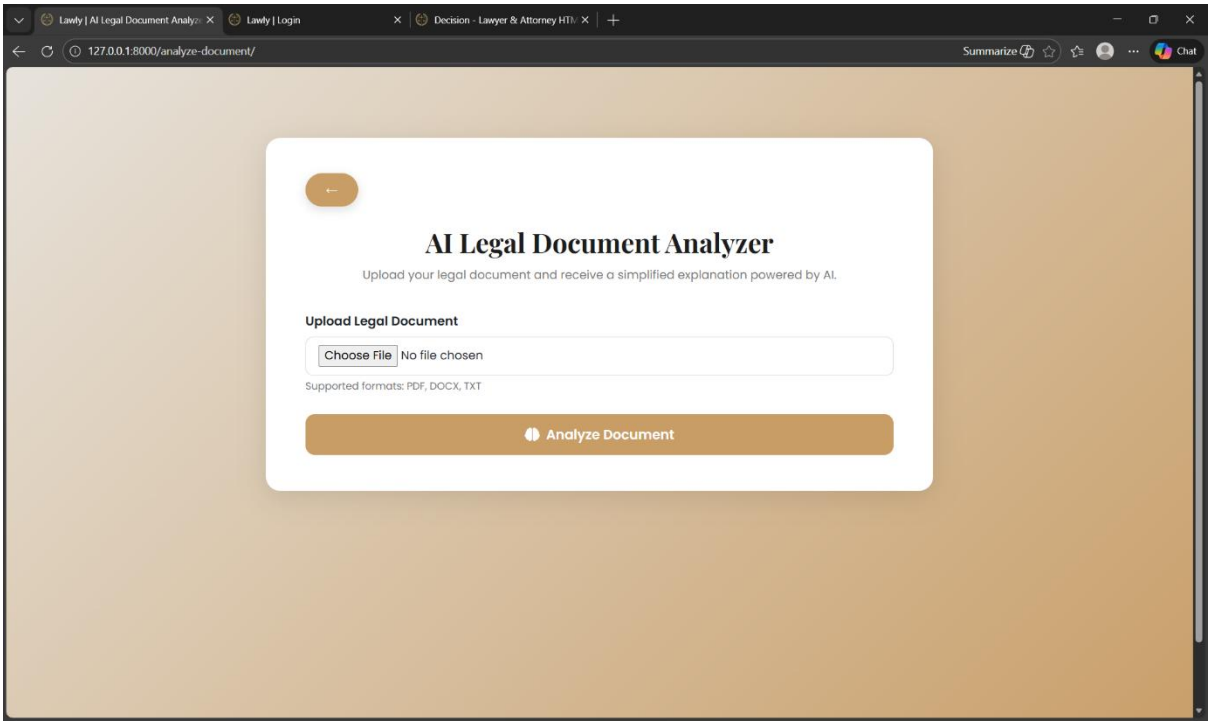

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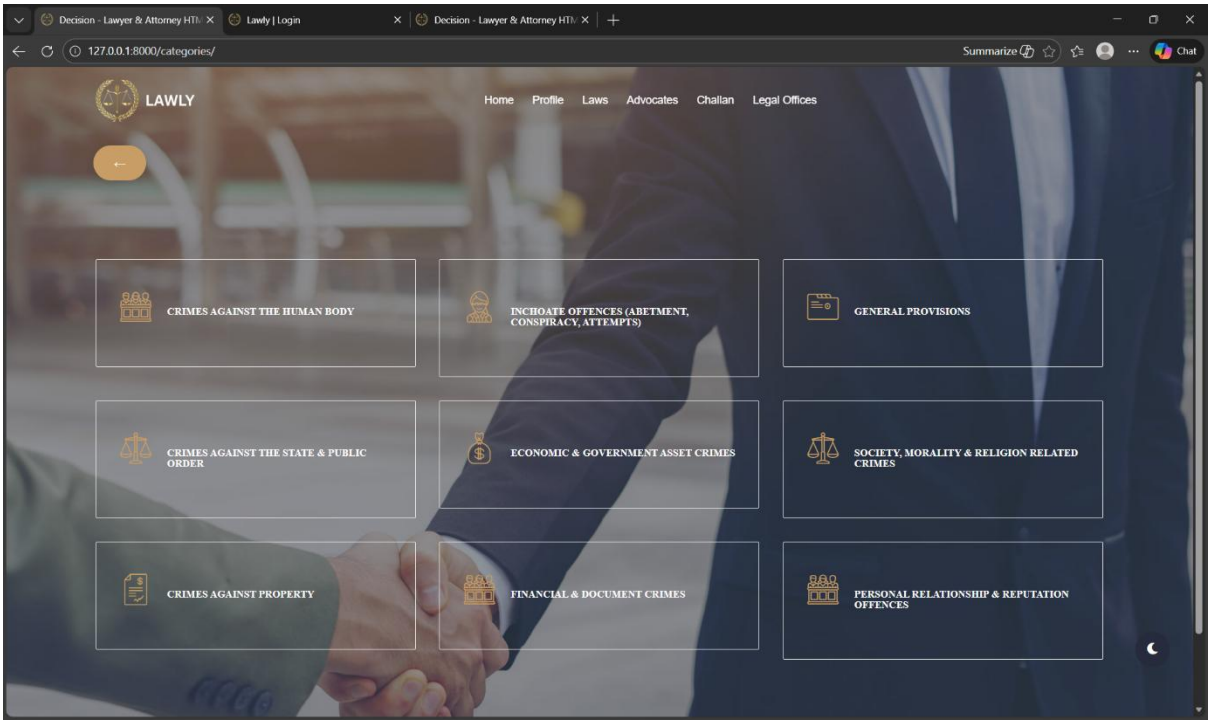

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